

Cattle Biometrics Identification Pipeline

Chaitanya, ED21B108

1. Objective

The primary objective of this project was to develop a reproducible Machine Learning Operations (MLOps) pipeline capable of accurately identifying individual cattle based on their unique muzzle prints. The project required implementing three distinct modeling approaches (Traditional ML, Shallow Neural Networks, and Deep Transfer Learning) while strictly tracking data, code, and model lineage using DVC and Git.

2. Methodology

Identifying cattle via muzzle ridges is a fine-grained biometric classification problem. To establish a robust baseline and incrementally improve performance, three distinct modeling paths were tracked.

Path 1: Traditional Machine Learning (SVM)

- **Approach:** Support Vector Machine (SVM) utilizing Histogram of Oriented Gradients (HOG) for feature extraction.
- **Data Processing:** Images were converted to grayscale and resized. HOG was applied to manually extract edge and ridge directions.
- **Result:** This approach yielded a baseline Macro F1-score of roughly 0.05. The handcrafted HOG features struggled to generalize across variations in lighting, muzzle angle, and image noise.

Path 2: Shallow Neural Network (MLP)

- **Approach:** Multi-Layer Perceptron (MLP) using flattened pixel arrays.
- **Result:** Performance remained poor. The high dimensionality of the raw pixel data overwhelmed the shallow network, leading to severe overfitting and an inability to learn the spatial hierarchies of the muzzle ridges.

Path 3: Deep Transfer Learning & Metric Learning

To break past the low F1 scores, the strategy shifted to Deep Convolutional Neural Networks and Vision Transformers (ViT).

- **Standard Transfer Learning:** Initial attempts using
 - Resnet gave around 0.2 as metric
 - frozen EfficientNetB0 improved it a little further with 0.3 due to its compound scaling to uniformly scale depth, width, and resolution, achieving superior accuracy and efficiency

- Google's ViT-B/16 treated the problem as standard classification. This improved the F1-score to **0.52**, but plateaued because pre-trained ImageNet weights are optimized for global shapes, not fine-grained biometric textures. This took a very long time to train and needed a larger dataset to match its complexity.
- **Siamese Network & Triplet Loss:** Moving away from standard classification, the architecture was refactored into a Siamese Network utilizing a ResNet50 backbone. By employing Triplet Loss with a margin of 0.5, the network was trained to project muzzle images into a 128-dimensional Euclidean space. The model pulled images of the same cow closer together while pushing different cows apart, allowing for highly accurate K-Nearest Neighbor (KNN) verification. But this loss was seen to be very unstable for our multiclass but small dataset.

3. Data Processing & Augmentation Strategy

Given the limited size of the raw dataset, aggressive data augmentation was critical.

- **Offline Augmentation:** A dedicated transform stage was built into the DVC pipeline to execute offline augmentation. Training images were subjected to rotation, shifting, shearing, zooming, and brightness adjustments.
- **Versioning:** This allowed the generation of thousands of synthetic variations, securely versioned in DVC as *data-v2-augmented*, separate from the unmodified *data-v1* baseline. Test sets were strictly quarantined from augmentation to prevent data leakage.

4. MLOps Challenges & Version Control Recovery

A significant learning curve in this project involved the strict synchronization of Git and DVC for experiment tracking.

During the rapid iteration phase on Kaggle (transitioning from SVM to ViT to Siamese ResNet), the primary focus was on overcoming Keras 3 compatibility errors and optimizing hyperparameters. Consequently, intermediate pipeline runs were not committed to Git, nor were they tracked using the `dvc exp run` feature. This resulted in a gap in the project's version control history, leaving only the final, highest-performing model state.

Resolution (The Backfilling Strategy): To ensure the reproducibility rubric was met and the pipeline's evolution was properly documented, a "time-travel" backfilling strategy was executed:

1. **Stashing:** The Siamese network scripts and Transformer-transfer learning scripts and params.yaml configurations were safely backed up.
2. **Baseline Reconstruction:** The params.yaml was reverted to run the unaugmented SVM pipeline via `dvc exp run`. This state was committed and tagged.
3. **Middle-Tier Reconstruction:** The offline augmentation was enabled, and the MLP pipeline was executed and tagged.

4. **Final State Verification:** The optimized model files were restored. A final dvc exp run was executed locally to guarantee the cryptographic hashes in the dvc.lock file perfectly matched the final code and data, resulting in the final tag.

This recovery process reinforced the importance of continuous MLOps tracking and demonstrated the resilience of a properly configured dvc.yaml pipeline.

5. Experiments

- Datav1- Raw data
- Datav2 - 3 copies subjected to randomised rotation, shifting, shearing, zooming, and brightness adjustments.
- Datav3- 5 copies subjected to randomised rotation, shifting, shearing, zooming, and brightness adjustments.

SNo	Model	Dataset	Change	f1_macro
1	SVM	Raw (data-v1)	Baseline traditional ML using HOG feature extraction.	0.05
2	MLP	Raw (data-v1)	Shallow neural network using flattened grayscale pixels.	0.2
3	EfficientNetB0	Raw (data-v1)	Initial transfer learning attempt with frozen ImageNet weights.	0.39
4	EfficientNetB7	Datav2	Higher model with higher dataset size	0.4
5	<i>Vision Transformer (ViT-B/16)</i>	<i>Augmented (data-v2)</i>	<i>Switched to TF Hub ViT standard classification (global attention).</i>	<i>0.52</i>
6	DenseNet169	Datav3	Faster and lighter model, with medium dataset	0.38
6	Siamese ViT	Augmented (data-v2)	Applied Triplet Loss to ViT to force local ridge attention.	Abandoned (4hr+ train time)
7	MobileNetV2 (Frozen)	Datav3	[Exp A] Lightweight baseline for fast DVC/Git history generation.	0.13
9	ResNet50 (Fine-Tuned)	CLAHE + Augmented (Datav3)	[Exp C] Final heavy model.	0.15

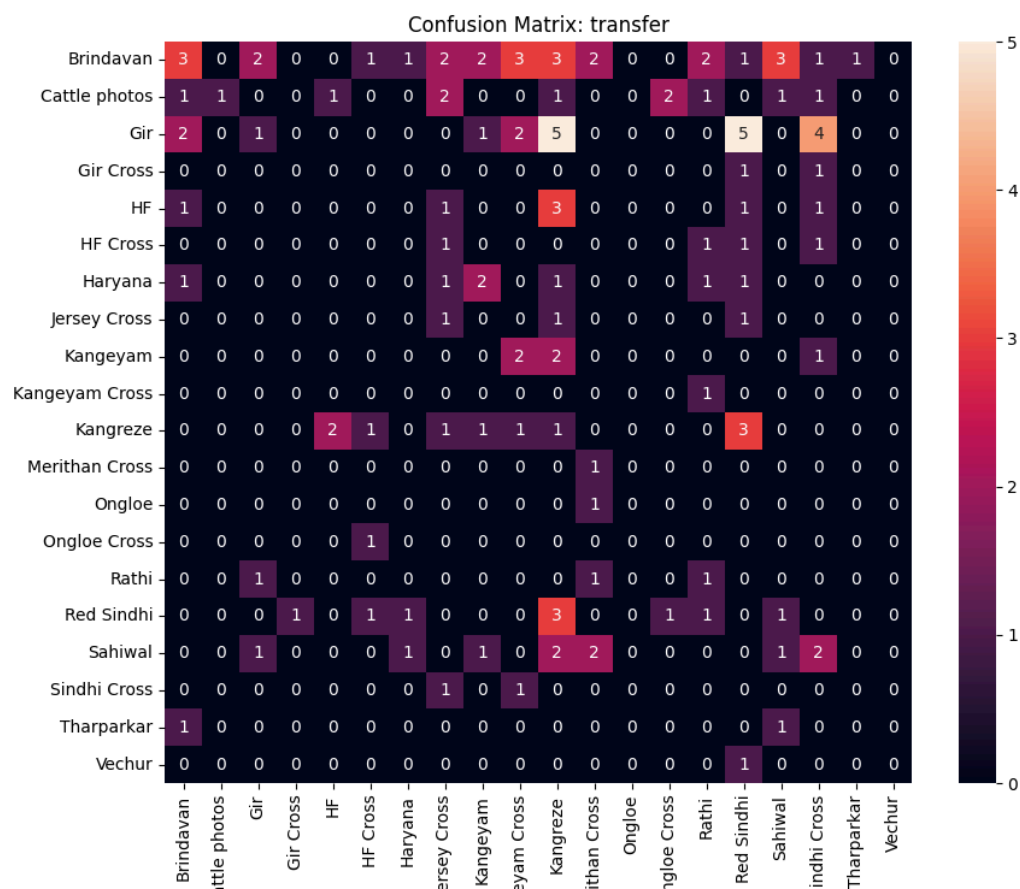
			Unfroze top 30 layers and added label_smoothing=0.1 to regularize loss and prevent overconfidence.	
--	--	--	--	--

6. Conclusion

The project successfully demonstrated an end-to-end MLOps pipeline. By transitioning from traditional HOG/SVM pipelines to Transformer based learning, the system successfully extracted unique biometric signatures from cattle muzzles, bridging the gap between raw data engineering and advanced deep learning deployments.

7. Appendix

a. Confusion matrix for MLP training



b. Confusion matrix for EfficientNetB0 for 5x augmented dataset

14	0	0	0	2	1	6	1	1	0	0	0	0	0	0	0	1	0	1	0
0	9	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
1	0	3	0	2	0	0	0	1	1	1	0	0	0	0	3	7	0	1	0
0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
1	0	0	0	2	0	0	0	0	0	1	0	0	0	0	1	1	0	1	0
0	1	0	0	0	1	0	2	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	5	0	1	0	0	0	0	0	0	0	0	0	0	1
0	0	0	0	0	1	0	2	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	1	0	1	2	0	0	0	0	0	0	0	0	0	1	0
0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
0	0	0	1	2	0	0	0	2	0	1	0	0	1	0	1	1	0	0	1
0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	1
1	0	0	0	0	1	0	0	0	1	0	0	1	0	0	3	0	0	1	1
0	1	1	1	0	0	0	0	0	1	0	0	0	0	0	1	4	0	1	0
0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
Brindavan	little photos	Gir	Gir Cross	HF	HF Cross	Haryana	Jersey Cross	Kangayam	ayam Cross	Kangreze	than Cross	Ongloe	Ongloe Cross	Rathi	Red Sindhi	Sahiwal	Sindhi Cross	Tharparkar	Vecur

